Visvesvaraya Technological University

BELAGAVI, KARNATAKA

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ

ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A PROJECT PHASE-1 REPORT

ON

"Product Originality Detector Using Blockchain"

Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

Submitted by

Rishabh Jayaram	**4JN22CS128**
Rishika	**4JN22CS129**
Rohith B S	**4JN22CS130**
Sanketh S T	**4JN22CS141**

Under the guidance of

Dr. Benakappa S M B.E., MTech., Ph.D.
Associate Professor, Dept. of CS&E,
JNNCE, Shivamogga.

Department of Computer Science Engineering

J.N.N College of Engineering

Shivamogga-577 204

MAY 2025

National Education Society ®

Jawaharlal Nehru New College of Engineering

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the project entitled

"Product Originality Detector Using Blockchain"

Submitted by

Rishabh Jayaram	**4JN22CS128**
Rishika	**4JN22CS129**
Rohith B S	**4JN22CS130**
Sanketh S T	**4JN22CS141**

Students of 6th semester B.E. CS&E, in partial fulfillment of the requirement for the award of degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-26.

Signature of Guide

Dr. Benakappa S M B.E, MTech, Ph.D.
Associate Professor, Dept. of CS&E

Signature of HOD

Dr. Jalesh Kumar MTech, Ph.D.
Professor and Head, Dept. of CS&E

ABSTRACT

In today’s global market, counterfeit products pose a significant threat to brand reputation, consumer safety, and economic integrity. The project presents an Originality Product Identification System leveraging Blockchain technology and QR codes to ensure the authenticity and traceability of products. The proposed system assigns each product a unique QR code, which is linked to a secure, tamper-proof record on a blockchain. At the time of manufacture, product data such as origin, manufacturer details, batch number, and timestamp are recorded on the blockchain. Consumers or stakeholders can scan the QR code using a smartphone application to instantly verify the product's originality and view its full history. The integration of blockchain guarantees data immutability and transparency, while QR codes provide a convenient and cost-effective interface for real-time verification. This system aims to combat counterfeiting, increase consumer trust, and improve supply chain transparency across various industries.

ACKNOWLEDGEMENT

On presenting the Project phase-1 report on **"Product Originality Detector Using Blockchain"**, we feel great to express our humble feelings of thanks to all those who have helped us directly or indirectly in the successful completion of the Project phase-1 report.

We would like to thank our respected guide **Dr. Benakappa S M,** Associate Professor, Dept. of CS & E, who has helped us a lot in making the project and for their continuous encouragement and guidance throughout the project work.

We would like to thank our respected coordinators **Mr. Vedananda D E**, Asst.Professor, **Mrs. Radhika S K**, Asst.Professor, Dept. of CS&E who have helped us in making the project and for their continuous encouragement and guidance throughout the project work.

We would like to thank **Dr. Jalesh Kumar**, Professor and Head, Dept. of CSE, JNNCE, Shivamogga and **Dr. Y Vijaya Kumar**, Principal, JNNCE, Shivamogga for all their support and encouragement.

We are grateful to the Department of Computer Science and Engineering and our institution Jawaharlal Nehru New College of Engineering for imparting us the knowledge with which we can do our best.

Finally, we also would like to thank the whole teaching and non-teaching staff of Computer Science and Engineering Department.

Thank you,

Rishabh Jayaram	**4JN22CS128**
Rishika	**4JN22CS129**
Rohith B S	**4JN22CS130**
Sanketh S T	**4JN22CS141**

Table of Contents

Abstract		*i*
Acknowledgement		*ii*
Chapter 1	**Introduction**	**1**
1.1	Problem Statement	2
1.2	Objectives	2
Chapter 2	**Literature Survey**	**3-9**
Chapter 3	**Summary of Literature Survey**	**10**
	References	**11-12**

Chapter 1

INTRODUCTION

Blockchain is the distributed system that is primarily used to store the record of the transactions that have taken uplace between two parties. Each of the transaction is a block, which linked in the sequence using cryptography, making it impossible to reverse it. This means that data validation in blockchain is distributed, unlike a centralized system of a traditional database. Smart contracts are self-executing programs residing on the blockchain that enforce the terms of the agreements between parties. This characteristic ensures that every activity within the blockchain network is irreparable and carried out by the system to enhance trust fewer digital transactions.

Piracy is a vice that goes round the world affecting the companies and consumers more so in such sectors as pharmaceuticals and fashion. Blockchain counters this by ensuring that products' information is immutable and can be accessed by anyone. Every product can receive a specific Quick Response (QR) code that will refer to its record on blockchain. If This code will also make it easier for the customers to verify whether they are using the genuine product by simply scanning the code with their phones. It enhances the confidence of the consumers and preserves the image of the brands in the market.

There are two major actors using the system: *manufacturers* and *consumers*. Smart Contracts help generate a one-time QR code on Ethereum for product details uploaded by manufacturers. Consumers read as it with a simple indicator interface through which they can get all the history of the product. It also protects data from alteration and, in case of reselling the product, smart contracts change the ownership on the blockchain. This helps to achieve the traceability of the products in the supply chain.

Blockchain technology is implemented to support smart contracts using Solidity and for development on the Ethereum platform, the Hardhat tool is used. Teams use website front-end development technology, which is React, while back-end technology is node.js. Ethers.js also is used for blockchain interaction. This multiple layered system not only provides security but also increases the level of supply chain visibility. This way it helps in reducing the losses that could be occasioned by counterfeit goods, enhances compliance to the law and increases customer trust. In the proposed solution blockchain is used to present a scalable approach to provide a check on product authenticity and supply chain.

PROBLEM STATEMENT AND OBJECTIVES

The counterfeit goods problem is escalating into a major world concern, compromising product security, brand confidence and national economy. In the current world, Supply Chain Management Systems (SCMS) and established authentication technologies like QR and Radio Frequency Identification Tags (RFID) are failing because they are susceptible to cloning, centralized control but lack transparency. It is because of these deficiencies that counterfeits are unlikely to be intercepted, therefore allowing them to enter the supply chain undetected. The urge calls for a secure, fool-proof, and almost flawless way of affirming the products from production straight to the last distribution, irrespective of the fact that products lose their value with time. Adoption of blockchain technology in counterfeiting prevention systems supports an immutable, verifiable, and decentralized system for recording transactions for the products. The application of a blockchain-based model for identifying counterfeits provides the supply chain players the ability to verify the integrity of the product at every stage.

1.1 Problem Statement

Counterfeiting is rising worldwide, negatively impacting product quality, brand reputation, and economies. Existing supply chain and authentication technologies such as QR codes and RFID have weaknesses like being easily cloned, centralized, and not transparent. These flaws make it difficult to effectively identify and prevent counterfeit goods. Hence, it is necessary to develop and use a secure and tamper-proof mechanism that enables authentication and traceability of products right from the producer all the way to the consumer. This is achieved through the creation of a decentralised blockchain-based anti-counterfeit product identification model that will help the various players in the chain to track and ensure that their products are legitimate, thus reducing on counterfeit products and hence enhancing consumer-confidence.

1.2 Objectives

1. To design and develop a blockchain-based system to authenticate products and prevent counterfeiting.
2. To implement QR codes and smart contracts to enable transparency and tamper-proof tracking of products throughout the supply chain.
3. To empower consumers to verify product authenticity by scanning QR codes linked to immutable blockchain records.
4. To enhance supply chain transparency and trust among stakeholders by providing a secure platform for product genuinety.

Chapter 2

LITERATURE SURVEY

Mass creation of counterfeit has become increasingly pressing concern to industries, consumers and economies in all parts of the world. However, this problem has called for various technologies to be introduced in Supply Chain Management (SCM) and product authentication, such as QR codes, RFID, and barcodes. The efficacy of these technologies to aid the prevention of the making of fake goods has been researched. By examining the existing body of research on these technologies, the present study identifies the strengths and weaknesses of these technologies while emphasizing the need for enhanced security and transparency in product authentication.

The authors [1], have proposed a Blockchain-based Management System (BCBCM) in which blockchain and QR codes are used to suppress counterfeiting of products in the chain of supply. Manufacturers produce QR codes containing encryption that protects product details that are cryptographically stored in the blockchain network. Through such QR codes consumers can validate the product and the system will check this information against blockchain log. When verification is failed, the system captures purchase data so as to identify the source of counterfeit item. Major elements of the system include blockchain, decentralized records, encryption measures, and identification of manufacturer in addition to utility of smartphones for scanning. Nevertheless, the strengths of the system are offset by prohibitively high upfront costs, the requirement for user training, scalability issues, concerns about data security, and the dependence on acceptance of mobile technology.

The authors [2], have described the deployment of a decentralized application (Dapp) on the Ethereum blockchain, a component of an Anti-Counterfeit Product System intended to mitigate the problem of counterfeit products. Manufacturers can advertise items online using encrypted, unchangeable labels through smart contract integration, (Secure Hash Algorithm) SHA-256 hashing and QR code technology. Consumers can freely access manufacturing and ownership records with just QR code scanning, which allows them to directly check legitimacy off the blockchain. Major issues such as registering companies, listing of products and monitoring ownership are all automated helped by smart contracts within the platform. This system enhances security, openness, trust, and efficiency, despite the difficulties which are connected with its complexity, low scalability, privacy risks, and requirement for online access.

In [3], they have described a framework which uses QR codes and cryptographic hashes to fight counterfeit products in supply chains. Manufacturers create individual QR codes along with a distributed ledger containing Ganache and MySQL, which enable retailers and end-users to verify the product within 600 second. The method provides secure, transparent, and self-verifiable data processes by interacting directly with the system instead of through third parties. This framework offers enhanced security, enhanced traceability, reduced dependence on central systems and flexibility to different sectors. Some of the challenges involve possible weaknesses in QR codes, lack of payment function, reliance on smartphones as well as strict time limits. Still, the system as such offers a working scheme, which can be supplemented by RFID and digital currency.

In [4], a decentralized Ethereum-based application that uses smart contracts and QR code infrastructures is proposed. The system identifies five different user roles, including administrator, manufacturer, supplier, retailer, and consumer who each have particular permissions that mandate consumers to authenticate goods by scanning QR codes. Through the use of Solidity, React, Node.js, Ethers.js, core Ethereum database PostgreSQL and being tested on Sepolia Testnet, the system delivers secure data management, shared governance, immediate tracking. In the case of dynamic gas fees that depend on full user participation, security flaws, scalability issues, and potential device compatibility obstacles, problems emerge. It is however realistic and creative solution to enhance supply chain integrity.

The authors in [5], describes a mechanism using temporary QR codes for verification of the authentic nature of the product. Blockchain based companies note their products' details and match them with distinctive QR codes, which then can be instantly verified by anyone scanning the codes. The platform uses smart contracts, Global Positioning System (GPS) data, and Firebase to run a decentralized database for product verification. The leading merits include enhanced security, improved consumer confidence, better brand protections, and more transparent corporate supply chain. Problems like significant first investments, the possibility of spoofs of QR codes, the need for consumer involvement and continued development of advanced technologies have to be addressed. Finally, it proposes a strong and flexible way to reduce counterfeits.

The authors in [6], makes use of blockchain technology and QR codes to detect counterfeit products. The latter consists of manufacturer, user, and cloud database apps which facilitate secure receipt of products, issuance of unique QR codes, and timely authenticating said products. Compared to the Consumer Food Price Index (CFPI), the study shows that Barcode Quality and Reader Performance Assessment (BQRPA) outperforms the study with superior scanner efficiency (up to 98. 86%), accuracy (up to 96.80 %) and performance (up to 98.79 %). Security is enhanced due to the immutability of the blockchain, supply chain is made transparent, product verification is done with ease through mobile apps and consumer trust is advanced through BQRPA. Still, the

system has to overcome problems such as QR forgery, huge costs upfront, need for online internet, and difficulties regarding manufacturer records administration. Finally, BQRPA provides a strong and flexible scale for product verification and the minimization of counterfeiting worldwide.

In [7], the authors propose that blockchain technology may be used to fight the counterfeiting problem in pharmaceuticals, electronics, and luxury goods. The proposed solution makes use of QR codes, Solidity smart contracts, and Ethereum in order to develop a trustworthy, transparent, and unchangeable system to authenticate products. The platform hashes various cryptographic tools such as SHA-256, Elliptic Curve Digital Signature Algorithm (ECDSA), and Keccak-256, and offers a simple-to-use understanding through the use of React, Node.js, and ethers.js to allow for quick Key benefits include increased transparency, increased consumer trust, immutability, and the scalability opportunity with the support of IoT and cross-chain prerogatives. Some of the vocal barriers include high first-cost, continuous need for web connection, limitation on privacy, and Ethereum's scalability edges.

The authors [8], describes a new product verification method that uses QR codes accompanied by (Advanced Encryption Standard) AES-256 encryption and SHA-256 hashing. Each product is accompanied by an individual QR code that has been encoded with the encrypted and hashed details, including product ID, date of its manufacture, and batch number. When the QR code is read on mobile application, information gets decrypted and authenticated by the system by matching the hash of the received message to the stored hash. The system receives 99.9% accuracy in identifying real products, 99.0% confidence with regard to reliability, and an average verification time of 0.8 seconds. Advantages include strong security, available-for-the-moment verification, low-cost implementation using open-source technology, and broad usability. Critical obstacles include making online connections, sustaining the high cost of establishing, and safe key operations. Still, it is a viable method to fight counterfeiting where it allows for offline implementation and gain of a broader market.

In [9], the system uses a unique QR codes and the Ethereum network. Supply chain manufacturers assign products QR codes and cryptography the codes into the blockchain using MetaMask, which enables suppliers to push updates. By using the QR code, users authenticate their purchases as the product's transaction record pops-up on the blockchain. Smart contracts are coded in Solidity while the integration with the blockchain is realized through Web3.js, the user interface uses React while testing is done through Ganache. Advantages include protection of safe, unmodified data, increased transparency in supply chain, decreased counterfeit sale, and financial efficiency. Although advantages are, issues remain with respect to scalability, transaction fee efficiency, technical infrastructure requirements, and the protection of user privacy. In a nutshell, this system is an open and decentralized approach to enhance the supply chain vulnerability.

In [10], a blockchain and QR code methodological approach in order to enhance tracking of products and curb counterfeiting in the supply circuit. Each product has a unique QR code containing its production information on an immovable blockchain. Stakeholders collaborate with the help of the QR codes to confirm and record information concerning products, and smart contracts also automate the process of confirmation with the least chances of errors. Consumers can verify physical products for authenticity in seconds because of an app available on mobile thereby creating confidence and transparency. By using Ethereum, smart contracts, and role-based access, the system enables manufacturers, retailers, and consumers to empower. Despite the benefits in traceability, counterfeit reduction and data security, the barriers are increased setup costs, requirement for technical solutions, potential vulnerability of QR code, scalability limits as well as privacy issues.

In [11], an Ethereum-decentralized system with smart contracts is recommended to support supply chain transparency and counteract the counterfeiting problem. This setup involves three major players i.e., the manufacturer, supplier and customer who work together through a Dapp to enter product information to the blockchain directly. Manufacturers create QR codes, suppliers authenticate the data and customers can verify the entire supply chain simply by reading the QR codes. The platform uses Ethereum, Ganache, Truffle, Web3.js, React, and MetaMask. Main benefits include product reliability, open record keeping, autonomous management of the supply chain and verification applied by consumers. However, the system faces resistance such as dependence on technology, high computational costs, the risk of QR code duplicates, and a need for a wide market approval.

The authors [12], have introduced a solution which utilizes single use scannable QR codes linked to a firebase database and can be accessed via a react.js interface. Each of the items is attributed with the individual QR code which ceases to function after one scan, and there is no chance of reuse. Firebase supports real-time data storage and authentication, analytics, and SHA secures the data from tampering. With the implementation of this system, it is easy for users to authenticate products and avert QR-code fraud, take advantage of smooth synchronization, process high volume and have easy user interface. However, challenges include leaning on network connectivity, scalability cost of Firebase, lack of blockchain security, and even some hesitance on the customers' side. It would be possible to further develop by integrating blockchain and IoT to boost traceability and automation provisions.

In [13], a thorough blockchain-based approach is presented in order to authenticate products and defeat the counterfeits using Hyperledger Fabric and Scratchable-QR codes. A Unique Identification (UID) is assigned to each item, thus allowing automatic creation, tracking and transfer of ownership with smart contracts and

Representational State Transfer Application Programming Interface (REST APIs). Stakeholders are now able to access current product data and monitor the entire supply chain in real time through “Hyper Supply”. Important benefits include enhanced security, increased consumer trust, better transparency, and protection from financial damage by counterfeit products. The push towards this system has a number of setbacks involving high costs, dependence on expert knowledge, a lack of adaption by the users, and size issues on scalability for small businesses. Despite these hindrances the system successfully protects the integrity of the supply chain.

The authors [14], presents a blockchain-backed system, which uses combination of Ethereum smart contracts and QR code scanning for counterfeiting tackling. Products move on a decentralized blockchain and are granted only a single, unique QR code, all of which enables consumers to verify authenticity with ease. Through hash chains, security is achieved, Key Performance Indicator (KPI) analysis and tools for user/admin provide performance monitoring and oversight. Prime benefits include stronger data security, transparent insight into supply chains, user-friendly capabilities, and more brand safeguarding. Confronted with challenges of reduced transaction capacities, operation costs, and protocols that have few adoptees, the system still overtakes conventional systems and offers a breeding ground for future integration with Artificial Intelligence (AI) and Internet of Things (IoT) practices.

The authors [15], describes the creation of a blockchain-based Production and Operations Management System (POMS) that should mitigate post-supply chain counterfeiting and RFID tag duplication problems. Built high on top of Ethereum, POMS provides every product with its distinct Engineering, Procurement, and Construction (EPC), which enables secure ownership verification and transfer with the use of smart contracts. It is integrated with RFID, QR, or NFC tags and has functionalities including enrollProduct(), shipProduct() and receiveProduct(). Core benefits are verifying consumers, decentralized control, and minimizing total operation costs. It is noteworthy that underlying problems originate from necessity of Ethereum, potential for privacy breaches, difficulties of changing ownership across several stakeholders, and rigidity of the protocol, all of which could overshadow a smooth interface. However, POMS provides a secure, and scalable manner for combating counterfeit goods.

The authors [16], discuss the problem of blockchain-based online reputation system fraud. It defines fraud based on fact or based on opinion and evaluates six threats that have the potential of differentiating these two including sybil and whitewashing. With associated real, blockchain-recorded transactions, and with the use of cryptographic privacy steps, blockchain shows success in minimizing actual fraud. Still, it is ineffective when it comes to subjective fraud as dishonest users will be able to use real transactions to hide artificial ratings.

Although blockchain increases trust and security levels, it is not effective enough to entirely eliminate rating fraud due to untrustworthy opinions and complex computations' necessity, which emphasizes the need for additional confirmation mechanisms.

The authors [17], uses a private blockchain to make product and customer info private before verification by QR codes or product ID. The system simplifies verification because of the incorporation of SHA-256 digital signatures and barcode recovery mechanism offering an increased time complexity of O(n), hence enhancing efficiency. It enjoys safe data handling, cost-effective transaction costs, better accuracy, and greater consumer trust because of unmodified records. Cloud storage allows the system to be scalable but key based access ensures the product's authenticity. However, its utilization is limited by the provision of a cap on 1,000 products, need for QR readers for its authentication, and possible issues with the performance of the cloud service. Federated Public Infrastructure and Smart Manufacturing Framework (FPISMF) provides a reliable, fast, and inexpensive way to combat counterfeit products with possible future advancements.

The authors [18], present the decentralized, Ethereum-based approach to solving counterfeiting issues in food and pharmaceuticals, for instance. Using a Dapp, products are registered, given unique labels and bound in Cryptographic QR Codes that give consumers power to authenticate goods and track ownership on the blockchain. With the help of smart contracts, ownership transfers on the blockchain become safe and inviolable, thereby creating more trust and transparency. These strengths include incorruptible data storage, high level of cryptography protection, intrinsic control without central authority and no third parties' participation in verification processes. On the negative side, these proposals suffer from the complexities of initial cost, complex technicalities, doubts on scalability and as well as complexities of using the system for non-exported users. Consequently, the system enhances reliability in product verification and creates a better consumer trust foundation.

The authors [19], proposes the development of a low-cost approach to help SMEs to fight counterfeits wherein the guidance is to apply Hashing, QR code technology and Web based system. Each product receives a unique ID and password, the hashed password is stored in a MySQL database, and a QR code is printed. To verify the products, a consumer only scans the QR code, inserts the ID in the manufacturer's website and a successful process of verification deactivates the record to ensure that it cannot be exploited in the future. This low-budget solution based on Hypertext Preprocessor (PHP), Hyper Text Markup Language (HTML), JavaScript, and qrlib.php produces quick response (40–94ms), convenient deployment, friendly interface, and annual costs ranging from $135. Although it has many good points, the centralized structure is liable to weakness like susceptibilities to database hacks and constant connection requirements. At the end of the day, the system

functions as a viable and cost saving approach to countering counterfeit goods in industries that run on thin margins.

In [20], the system developed is built on Ethereum blockchain, Solidity smart contracts and Web3.js, the BCPIS offers a decentralized solution to stop counterfeit goods in supply chain. This system allows for manufacturers to input the details of the products and to authenticate items through the individual QR codes used by customers in a secure, impenetrable, and transparent environment, all under Ganache during development. The system is determined with methods such as the use of Elliptic Curve Digital Signature Algorithm (ECDSA) for data encryption, using Proof of Work (PoW) towards consensus, and the use of modules customized towards ownership monitoring and transaction validation. Some major advantages of the system are improved security, open data, decentralized control over data, costs much lower than traditional forms of tags, such as RFID and holograms, and easy verification for customers. However, they can be confronted by such tasks as significant electricity usage, poor scalability, high initial cost, reduced interoperability, and growth in storage requirements. In spite of such issues, the system effectively enhances trust and accountability with respect to product authentication and ownership verification.

Chapter 3

Summary of Literature Survey

There is an increasing volume of scholarly pieces that refer to the increasing problem of product counterfeiting as a serious problem across industries, from pharmaceutical products, electronics to fashion and consumer goods. Counterfeit goods cause huge financial damage to organizations, and also pose a danger to consumer safety and brand image. Popular authentication and supply chain technologies of the likes of QR code, RFID and barcode have been deployed to track and authenticate goods. However, robust investigation has shown severe shortcomings in such technologies. Among the disadvantages that emerge in these studies include such problems as being prone to cloning, lack of visibility, and reliance on centralized databases to be hacked or fail. For these flaws, counterfeiters easily avoid these safeguards, and it becomes difficult for businesses and consumers to distinguish the real from the counterfeit goods.

To address these problems, many researchers have started using blockchain technology because of its decentralized, immutable, and transparent properties. Several studies reviewed here highlight that linking blockchain with smart contracts and QR codes gives each stakeholder the ability to confirm product authenticity throughout the supply chain. Implementations presented in the reports rely on Ethereum, Solidity, smart contracts, and use React, Node.js, and Web3.js on both ends. These technologies provide reassurance to consumers and enhance product visibility, yet implementation hurdles are substantial, including substantial costs, technical scalability difficulties, user acceptance problems, and the necessity for continuous internet availability.

Multiple reviewed works propose the integration of advanced cryptographic techniques, such as SHA-256 and AES-256, to increase the reliability of product information found in QR codes. Even with the replication of QR codes, the authenticity of products can be confirmed through these encryption techniques on the blockchain. Certain proposed solutions add features, including GPS location tracking, single-use QR codes, and transactions controlled by smart contracts, to further strengthen the accuracy and usefulness of anti-counterfeiting systems. Laboratory tests on these techniques show high rates of accuracy, user acceptance, and efficient workflow, justifying their use in real-world applications.

REFRENCES

[1] Jayaprasanna MC, Soundharya VA, Suhana M, Sujatha S. A block chain-based management system for detecting counterfeit product in supply chain. Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV) 2021 Feb 4 (pp. 253-257). IEEE [2021].

[2] Singhal I, Bisht HS, Sharma Y. Anti-counterfeit product system using blockchain technology. Int. J. Res. Appl. Sci. Eng. Technol. 2021 Dec;9(12):291-5.

[3] Prathipa S, Harish K, Thashanmouli N. Counterfeit product detection in supply chain management with blockchain. 1st International Conference on Computational Science and Technology (ICCST) Nov 9 (pp. 841-844). IEEE [2022].

[4] Lo YX, Tan CC. Blockchain-Based Anti-Counterfeit Product Identification System. In2023 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET) Sep 12 (pp. 200-205). IEEE [2023].

[5] Pragadeesh M, Vignesh BK, Ashraf SM, Kumar RD. Anti-Counterfeiting Solution for Product Authentication Using Blockchain. In2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE) Feb 22 (pp. 1-4). IEEE [2024].

[6] Geetha R, Jayakumar D, Prabakaran P, Nivedha S, Sindhu M, Thivyarathi RC. An Experimental Evaluation of an Effective QR Code Based Duplicate Product Detection Using Blockchain Technology. In2024 International Conference on Intelligent Systems for Cybersecurity (ISCS) May 3 (pp. 1-7) IEEE [2024].

[7] Dhore ML, Loya R, Sasne R, Rokade T, Chandak R. Anti-Counterfeit Product Identification System: A Paradigm Shift in Supply Chain Integrity. In2024 8th International Conference on Computing, Communication, Control and Automation (ICCUBEA) 2024 Aug 23 (pp. 1-6). IEEE [2024].

[8] Shankar BP, Rajkumar N, Viji C, Kumar KD, Saravanakumar S, Mohanraj A. Approach to Identifying Counterfeit Products with QR Codes and Computational Algorithms. In2024 5th International Conference on Smart Electronics and Communication (ICOSEC) Sep 18 (pp. 1203-1209). IEEE [2024].

[9] V. Mane, R. Patil, S. Pawar, and N. Pujari, “Detection of Counterfeit Products with QR Codes using Blockchain Technology,” in 2024 International Conference on Circuit Power and Computing Technologies (ICCPCT), 2024, pp. 545–550.

[10] Keerthy G., Neha Rohith, Hemanth B.G., Amay Deepak Nayak, and 5D B. Srinivas, “Enhancing Product Authentication and Counterfeit Detection Using QR Codes and Blockchain Technology,” Proceedings of the Second International Conference on Edge Computing and Applications (ICECAA 2023), IEEE, pp. 262–267, 2023.

[11] K. Wasnik, Sondawle R. Wani, and N. Pulgam, "Detection of Counterfeit Products using Blockchain," ITM Web of Conferences, vol. 44, pp. 03015, 2022.

[12] Jhansi Devi P, Gutti D. N, Dutta M. S, Domakuntla S. K & Damerakonda M. (2023). One Time QR-Code for Fake Product Identification. Proceedings of the 5th International Conference on Inventive Research in Computing Applications (ICIRCA 2023), IEEE. ISBN: 979-8-3503-2142-5.

[13] M. A. Muzafar, A. Bhargava, A. Jha, and P. Nand, "Securing the Supply Chain: A Comprehensive Solution with Blockchain Technology and QR-Based Anti-Counterfeit Mechanism," International Journal of Performability Engineering, vol. 19, no. 5, pp. 312–323, May 2023.

[14] D. M. R, S. A. M. G Shukur, and M. G. Shukur, "Securing Authenticity: A Frontier Blockchain-Powered Fake Product Detection via QR Codes," 2024 Second International Conference on Advanced Computing & Communication Technologies (ICACC Tech), pp. 188–194, 2024.

[15] K. Toyoda, P. T Mathiopoulos, I. Sasase, and T. Ohtsuki "A Novel Blockchain-Based Product Ownership Management System (POMS) for Anti-Counterfeits in the Post Supply Chain," IEEE Access, vol. 5, pp. 17465–17477, 2017.

[16] Cai Y, & Zhu D. (2016). Fraud detections for online businesses: A perspective from blockchain technology. Financial Innovation, 2(1), 20.

[17] Gupta S, Kuchipudi R, Sohail M, Singh K, Mahalakshmi J & Sarabu A. (2024). Fake product identification for small and medium firms (FPISMF) using blockchain technology. Measurement: Sensors, 33, 101164.

[18] Singhal I, Bisht H. S, & Sharma Y. (2021). Anti-Counterfeit Product System Using Blockchain Technology. International Journal for Research in Applied Science and Engineering Technology (IJRASET), 9(12).

[19] Hossain M, Begum M, Das B. C & Uddin J. (2024). A cost-effective counterfeiting prevention method using hashing, QR code, and website. International Journal of Advances in Applied Sciences (IJAAS), 13(2), 351–359.

[20] Mhatre M, Kashid H, Jain T & Chavan P. (2022). BCPIS: Blockchain-based Counterfeit Product Identification System. Journal of Applied Security Research.